Regression Model Selection

Choose the model based on the outcome type, estimand, data structure, and functional form—then interpret it on the scale that matters.

START WITH THE QUESTION

OUTCOME
What is measured?

→

GOAL
What job?

→

STRUCTURE
Where is dependence?

Outcome + Goal + Structure

OUTCOME

Continuous · binary · count · ordinal · nominal · time-to-event · repeated / clustered

GOAL

Prediction · explanation · causal estimation · adjustment · subgroup comparison

ESTIMAND

Mean difference · probability difference · odds ratio · rate ratio · expected count

CORE PRINCIPLE

Match the model to the question—and make the estimand explicit before fitting.

VISUAL INTERVIEW & LEARNING CHEAT SHEET · POSTER 2

Model selection map

Match the model family to the outcome-generating process.

OUTCOME → COMMON MODEL

Outcome	Common model
Continuous	Linear regression
Binary	Logistic regression
Count	Poisson / Negative Binomial
Ordinal	Ordinal logistic
Nominal multicategory	Multinomial logistic
Time-to-event	Survival model
Repeated / nested	Mixed / multilevel model

THEN REFINE

- ✓ estimand and interpretation
- ✓ functional form
- ✓ dispersion / zero process
- ✓ dependence
- ✓ prediction vs causality

MEMORY HOOK

Outcome family first.
Structure and estimand next.

Linear regression

Use for a continuous outcome when the conditional mean can be represented reasonably well.

MODEL

$$Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon$$

COEFFICIENT

$$\beta_j$$

Expected change in Y for a one-unit change in X_j, holding modeled covariates constant.

MAIN ASSUMPTIONS

- ✓ appropriate functional form
- ✓ independent errors at the relevant level
- ✓ reasonable variance structure
- ✓ no perfect multicollinearity

NORMALITY BOUNDARY

Raw Y need not be normal.

The model concerns the conditional outcome structure and residual behavior—not whether the outcome itself looks perfectly normal.

INTERVIEW MOVE

Diagnose residuals and functional form; do not reject linear regression only because the raw outcome is skewed.

Logistic regression

For a binary outcome, model log-odds—but communicate on a scale people understand.

BINARY OUTCOME

$$Y \in \{0,1\}$$

$$\log[p/(1-p)] = \beta_0 + \beta_1 X_1 + \cdots$$

where p = P(Y=1 | X)

EXPONENTIATED COEFFICIENT

$$e^{\beta_j} = \text{odds ratio}$$

An odds ratio is not a risk ratio or probability difference.

STAKEHOLDER SCALE

- ✓ predicted probability
- ✓ marginal effect
- ✓ risk difference
- ✓ odds ratio when appropriate

MEMORY HOOK

Logistic models log-odds;
people usually think in probabilities.

Count models: Poisson or Negative Binomial

Use count models for sessions, purchases, errors, contacts, and other nonnegative event totals.

POISSON

$$E[Y|X] \approx Var(Y|X)$$

$$\log E[Y|X] = X\beta$$

Exponentiated coefficients can be interpreted as rate ratios.

NEGATIVE BINOMIAL

$$Var(Y|X) > E[Y|X]$$

Useful when substantial overdispersion remains beyond the Poisson variance.

DIAGNOSE

- ✓ conditional dispersion
- ✓ exposure time / offset
- ✓ zero process
- ✓ dependence
- ✓ tail behavior

MEMORY HOOK

Poisson = mean roughly variance.
Negative Binomial = extra count variability.

Many zeros: hurdle vs zero-inflated

Do not choose a special zero model merely because zeros are common. Ask why they occur.

HURDLE MODEL

Two stages

Any occurrence? P(Y>0) → **How much? Y | Y>0**

Useful when crossing from zero to positive is substantively distinct from intensity.

ZERO-INFLATED MODEL

Two sources of zero

- structural zero
- ordinary count-process zero

SELECTION QUESTION

Why are the zeros occurring?

Mechanism comes before specialized likelihood.

BOUNDARY

Use a zero-inflated model only when the two-process story makes substantive sense.

VISUAL INTERVIEW & LEARNING CHEAT SHEET · POSTER 7

Ordinal & multinomial outcomes

Ordering determines the model family and the meaning of coefficients.

ORDINAL OUTCOME

Low < Medium < High

Ordinal logistic

A common assumption is proportional odds.

NOMINAL OUTCOME

A, B, C

Multinomial logistic

Categories are distinct but unordered.

MODEL CHECK

- ✓ is ordering defensible?
- ✓ does proportional odds fit?
- ✓ is the reference category clear?
- ✓ are probabilities interpretable?

MEMORY HOOK

Ordered → ordinal.
Unordered → multinomial.

Model nonlinearity directly

A continuous predictor does not have to be forced into one straight-line slope.

FLEXIBLE FUNCTIONAL FORM

$$Y = \beta_0 + f(X) + \varepsilon$$

Transformations → Polynomial terms → Splines → Piecewise → GAM

WHY IT MATTERS

Misspecified shape can distort predictions, residuals, and the interpretation of other coefficients.

PREFERRED APPROACH

Keep the continuous information.

Model the shape, inspect it, and summarize meaningful contrasts.

AVOID

Arbitrary categorization

Cutting a continuous variable into bins loses information and creates threshold artifacts.

KEY PRINCIPLE

Do not discretize a continuous predictor just because its relationship is nonlinear.

Interactions = differences in effects

Use an interaction when the effect of one predictor depends on another.

MODEL

$$Y = \beta_0 + \beta_1 X + \beta_2 Z + \beta_3 XZ + \varepsilon$$

CONDITIONAL EFFECT OF X

$$\beta_1 + \beta_3 Z$$

The effect must be interpreted at specified values of Z.

COMMON MISTAKE

“Significant in one subgroup, not significant in another” does not itself establish a significant difference.

WHAT B_3 TESTS

Do effects differ?

It is a test of the difference between effects.

MEMORY HOOK

Interaction = Difference in Effects

Multicollinearity

Correlated predictors can destabilize individual coefficients even when predictions remain useful.

SYMPTOMS

- ✓ large standard errors
- ✓ coefficient sign changes
- ✓ sensitivity to specification

Diagnostic: VIF

WHAT IT MAINLY AFFECTS

Interpretation + precision

Prediction may remain good because correlated predictors carry similar combined signal.

POSSIBLE RESPONSES

- ✓ combine redundant variables
- ✓ remove one when substantively justified
- ✓ use ridge regression
- ✓ de-emphasize individual coefficients for prediction

MEMORY HOOK

Predictions can be stable while coefficients are unstable.

Regularization controls complexity

Useful with many or correlated predictors, high overfitting risk, and prediction-focused goals.

RIDGE · L2

Loss + $\lambda\Sigma\beta_j^2$

Shrinks coefficients toward zero; helpful for correlated predictors.

LASSO · L1

Loss + $\lambda\Sigma|\beta_j|$

Can set coefficients exactly to zero; useful for sparse prediction.

ELASTIC NET

Ridge + Lasso

Useful when many predictors form correlated groups.

TUNE HONESTLY

Choose λ by validation / cross-validation

All preprocessing and tuning belong inside the validation pipeline.

BOUNDARY

Detailed validation strategy belongs in the **Model Evaluation & Validation** sheet.

Dimension reduction: PCA

PCA compresses correlated predictors into fewer components.

TRANSFORMATION

Original predictors → **Orthogonal components** → **Smaller representation**

USEFUL WHEN

- ✓ dimensionality is high
- ✓ predictors are strongly correlated
- ✓ prediction matters more than original-variable interpretation

TRADE-OFF

Compression costs interpretability.

Components may be harder to explain than the original variables.

MEMORY HOOK

Regularization shrinks.
PCA compresses.

Variable selection follows the goal

Purely mechanical significance screening can create unstable and biased models.

AVOID

“Keep only variables with p < .05.”

- unstable models
- biased coefficients
- poor reproducibility
- omitted-variable problems

CAUSAL / EXPLANATORY

Identification

Theory · DAGs · confounding structure · pre-treatment status

PREDICTION

Generalization

Validation · regularization · out-of-sample performance

MEMORY HOOK

Causal selection → Identification
Predictive selection → Generalization

Dependence & multilevel structure

Independent-error regression can underestimate uncertainty when observations are repeated or nested.

WHERE DEPENDENCE LIVES

Measurements → **within Users** → **within Markets**

Where does dependence live?

POSSIBLE APPROACHES

- ✓ cluster-robust standard errors
- ✓ fixed effects
- ✓ mixed / multilevel models
- ✓ GEE

USE MULTILEVEL MODELS TO MODEL

- ✓ group heterogeneity
- ✓ partial pooling
- ✓ random intercepts
- ✓ random slopes

SELECTION PRINCIPLE

Choose based on the estimand, dependence structure, number of clusters, and whether group effects themselves matter.

Robust standard errors: useful but limited

Heteroskedasticity-robust SEs can improve uncertainty estimation when the variance model is misspecified.

WHAT THEY CAN IMPROVE

Some standard-error estimates

They can protect inference against certain forms of heteroskedasticity.

WHAT THEY DO NOT FIX

- nonlinear functional form
- omitted confounding
- selection bias
- measurement error
- incorrect outcome family

MEMORY HOOK

Robust SEs fix some uncertainty estimates—not the underlying model.

Interpret on a meaningful scale

A technically correct coefficient is not automatically a useful explanation.

LINEAR MODEL

Report ΔY

Translate a realistic change in X into expected outcome units.

LOGISTIC MODEL

Predicted probability · marginal effect · risk difference · odds ratio when appropriate

COUNT MODEL

Expected count · rate ratio · change over a meaningful exposure period

INTERACTION

Show predicted outcomes or contrasts across meaningful moderator values.

KEY PRINCIPLE

Translate coefficients into quantities stakeholders can reason about.

COMMUNICATION PIPELINE

Coefficient → **Predicted quantity** → **Meaningful contrast** → **Decision implication**

Common interview traps → better answers

State the limitation, then connect the correction to outcome, estimand, structure, or modeling goal.

EIGHT TRAPS

× "The outcome is skewed, so linear regression is invalid."
I care about the conditional mean structure and residual behavior, not whether the raw outcome itself is perfectly normal.

× "Binary outcomes can only be communicated as odds ratios."
Logistic regression models log-odds, but predicted probabilities or marginal effects may be easier to interpret.

× "Poisson is appropriate for any count outcome."
I would inspect dispersion and the data-generating process.

× "Many zeros means use a zero-inflated model."
I need a substantive reason to believe there are two zero-generating processes.

× "A significant interaction means treatment works in one group but not another."
The interaction tests whether the effects differ.

× "High VIF means the whole model is useless."
It mainly warns that individual coefficients may be unstable.

× "Choose variables using p-values."
Variable selection should reflect the goal—identification for causal work, generalization for prediction.

× "Robust standard errors fix misspecification."
They improve some SE estimates but do not fix functional form or confounding.

Regression model selection flow

Move from the data-generating question to an interpretable and honestly validated model.

ELEVEN-STEP FLOW

1 · Outcome

2 · Goal

3 · Data structure

4 · Model family

5 · Functional form

6 · Covariates by goal

7 · Control complexity

8 · Account for dependence

9 · Diagnose structure

10 · Interpret meaningfully

11 · Validate prediction

FAMILY MAP

Continuous → Linear
Binary → Logistic
Count → Poisson / NB
Ordinal → Ordinal logistic
Repeated → Multilevel / correlated data

SHAPE MAP

Linear? · Nonlinear? · Interaction?

GOAL MAP

Causal → identification
Prediction → generalization

VALIDATION HANDOFF

Validate predictive models using the Model Evaluation & Validation framework.

30-second interview answer

Lead with outcome, estimand, and structure; then explain shape, covariates, complexity, interpretation, and validation.

READY-TO-USE ANSWER

"I'd choose the regression model based first on the outcome, the estimand, and the structure of the data. For continuous outcomes I might start with linear regression, for binary outcomes logistic regression, and for counts Poisson or Negative Binomial depending on dispersion. I'd then check whether the functional form is appropriate, whether interactions or nonlinear terms are needed, and whether clustering or repeated measures require a multilevel or correlated-data approach. Covariate selection would depend on the goal: causal identification for explanatory work versus generalization for prediction. If the model is high-dimensional, I might use regularization or dimension reduction. Finally, I'd interpret the result on a meaningful scale and validate predictive models using a separate evaluation framework."

ANSWER ARCHITECTURE

Outcome + estimand → Family → Shape + interaction → Dependence → Goal-based covariates → Interpret + validate

STRONG CLOSING

The model is defensible when its family, functional form, dependence assumptions, and interpretation all match the question.

VISUAL INTERVIEW & LEARNING CHEAT SHEET · POSTER 20

Regression selection recall board

Use these hooks to reconstruct the full interview answer under pressure.

START

Outcome + Goal + Structure

FUNCTIONAL FORM

Linear? · Nonlinear? · Interaction?

VARIABLE CHOICE

Causal → Identification

Prediction → Generalization

COMPLEXITY

Regularize | Reduce | Simplify

COUNTS

Poisson: mean ≈ variance
NB: extra variability

MEANINGFUL SCALE

Coefficient → Predicted quantity → Decision

MOST IMPORTANT RULE

Choose the regression model to match the outcome, estimand, and data structure–then evaluate it using the right validation framework.